Farbod Haeri

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Summary

Mechanical engineering student (M.S., UC San Diego) specializing in controls and robotics. I design and build the hardware and write the controls and software that make it behave, most recently a fully-actuated hexacopter I took from CAD to 6-DoF flight, tuned in MATLAB and validated in motion capture. What pulls me is the control side: the modeling and tuning that turn a pile of parts into a machine that holds steady and does what it's told.

Skills

Programming: Python, MATLAB, C++, Java, JavaScript, HTML/CSS, Flask, Git
CAD / CAM / CAE: SolidWorks (Simulation, PDM), Fusion 360, NX, AutoCAD, ANSYS (FEA)
Manufacturing: CNC mill, manual lathe, waterjet, laser cutter, FDM 3D printing
Electronics: wiring, power distribution, motor / ESC integration, flight-controller (Pixhawk) bring-up, soldering (through-hole + SMD)
Controls: PID, lead-lag, notch filtering, state-space / LQR, loop shaping, pole placement, sensor fusion
Test & Metrology: optical metrology, machine vision, fatigue testing, defect quantification, dataset auditing

Research Experience

Undergraduate Researcher, UAV Controls & Robotics — Jan 2026 – Present
Prof. Thomas Bewley · UC San Diego — *San Diego, CA*

- Designing and building a fixed-tilt hexacopter, fully actuated in six degrees of freedom: its rotors sit at fixed cant angles, so it slides in any direction and holds a steady attitude instead of tilting to move like a normal multirotor. Built for target tracking and tight-space search.
- Own the build end to end: designed the airframe in SolidWorks, printed it as a single 3D-printed monocoque to keep it light and stiff, and assembled two flying prototypes.
- Ran thrust-stand tests comparing ducted and un-ducted propellers, measuring thrust, torque, and efficiency, and used the data to settle the propulsion and ducting instead of guessing at it.
- Designing the 6-DoF flight controller that decouples translation from rotation; deriving and tuning the gains in MATLAB, then checking attitude and position control against motion-capture ground truth.

Undergraduate Researcher, Advanced Materials & Metrology — Aug 2025 – Dec 2025
Prof. Maziar Ghazinejad · UC San Diego — *San Diego, CA*

- Built MATLAB pipelines turning raw optical-metrology and machine-vision images into defect-fraction and surface-roughness data, feeding clean, repeatable numbers into fatigue-life models for additively-manufactured steels.
- Designed 10+ mechanical components and machined the fatigue and metrology fixtures myself on the CNC mill and waterjet.
- Ran fatigue tests across material variants, logged the stress data, and audited the datasets so every result stayed traceable.

Projects

Automated Cognitive-Training System for Mice | *Senior Design, Jared Young Lab, UCSD* — Jan 2026 – Jun 2026

- Built the operator dashboard (HTML / CSS / JavaScript) the lab uses to launch sessions and watch live per-mouse reaction time, accuracy, and performance across multiple cages.
- Wrote the Python / Flask supervisory software, a training-cycle state machine and per-mouse scheduler that runs every stage from habituation through the 5-choice task, plus the data-logging and the Wi-Fi / MQTT link to the touchscreen rigs, so protocols change in code instead of re-flashing hardware.

MAE3 Autonomous Robot | *1st Place, UCSD MAE3 Robot Competition* — Sep 2024 – Dec 2024

- Led mechanical design on a small team; built the chassis and drivetrain from scratch and iterated through three revisions to take first place.

Work Experience

Student Engineer Intern — May 2024 – Aug 2024
County of Sacramento — *Sacramento, CA*

- Benchmarked energy use across 120+ county buildings, scoring each building's efficiency and rebuilding 50 buildings' utility histories into clean baselines to flag the low performers for audit and energy reduction.

Materials Manager — Jun 2022 – Apr 2024
Apple — *Elk Grove, CA*

- Planned and kitted inventory across 500+ SKUs for new-product builds, flashed and tested production units, and escalated supply risks to keep lines from stopping.

Education

University of California, San Diego — San Diego, CA
M.S. Mechanical Engineering — 2026 – 2027
B.S. Mechanical Engineering, Specialization in Robotics & Controls — 2024 – 2026